

Why does Multilingual Training improve Theorem Proving?

A Small-Scale Controlled Study

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CMSC848T

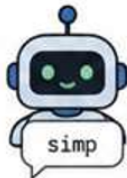
Neural Theorem Proving | Proof Search | Multilingual Training

1 What is theorem proving?

1 Proof state

Goal:
 $\forall n : \mathbb{N}, n + 0 = n$

2 Model suggests
a tactic



3 Proof assistant
checks it



The model sees the current proof situation, suggests a next step, and the proof assistant checks whether it works.

ProofWala result:

Multilingual training improves it!

Research Question

Why does multilingual training improve theorem proving performance?

H1: Cross-language Transfer

Models learn shared semantic patterns across proof systems.



Semantics

VS

H2: Regularization

More varied syntactic training data improves robustness.



Syntax

Our Training Experiment

Three Controlled Training Setups

Model	Training Data	Purpose
E1	Lean only	Baseline
E3	Lean + Rocq	Real multilingual
E4	Lean + pseudo-Lean variant	Regularization control

E1: Lean only



keeps Lean
semantics & syntax

E3: Lean + Rocq



combines Lean and
Rocq semantics &
syntax

E4: Lean + pseudo-Lean



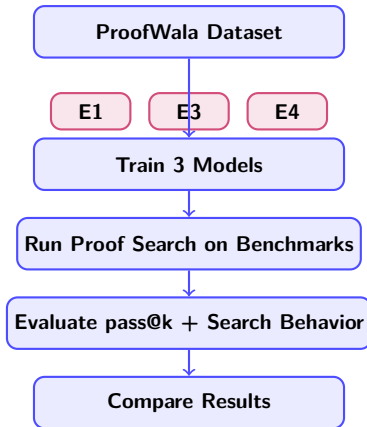
```
h -> pseudo_1  
x -> pseudo_2
```

keeps Lean semantics
but **varies** Lean syntax

Methodology

Shared Setup

- ▶ Same architecture: CodeT5-small
- ▶ Same tokenizer
- ▶ Same number training steps
- ▶ Same proof-search algorithm
- ▶ Same evaluation benchmarks



Benchmarks:

Easy_Lean

Set of easy-to-prove Lean theorems whose proofs share similar semantic patterns

Hard_Lean

Set of harder-to-prove Lean theorems with more diverse semantic proof patterns

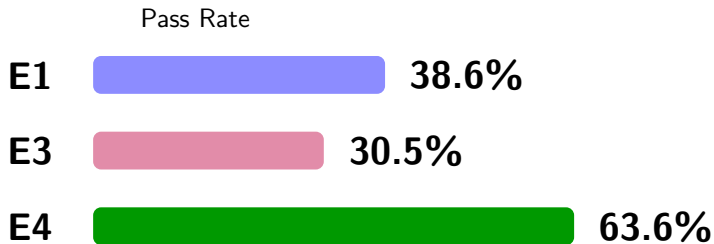
Key Experimental Design

We changed only the training data.

This isolates the effect of multilingual vs. pseudo-multilingual training and strengthens causal interpretation.

Main Results

Easy_Lean Results



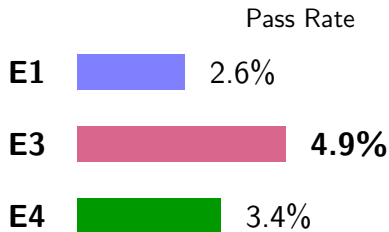
E4 $\gg$ E1 $>$ E3

Interpretation

Only-syntax variation helped more than syntax variation & combined semantics on problems whose solutions have similar combinations of steps!

Main Results

Hard_Lean Results



E3 > E4 > E1

Interpretation

Syntax variation & combined semantics may help more than only-syntax variation on problems that require solutions with fundamentally different ideas.

Main Interpretation

Our benchmarks reveal different mechanisms

Easy_Lean

E4 wins

Pseudo-multilingual



H2 is supported **when**
solutions have similar
combinations of steps

Hard_Lean

E3 wins

Real multilingual



H1 is supported **when**
solutions require
fundamentally different ideas

Main Conclusion

H1 and H2 hold under different circumstances.

Key Findings

Main Experimental Findings

- ▶ Multilingual gains are not driven by a single mechanism
- ▶ Pseudo-multilingual training (i.e., only-syntax variation) can improve performance on theorems whose proofs have similar combinations of tactics.
- ▶ Real multilingual training (i.e., syntax variation and combined semantics) may improve performance on theorems whose proofs require newer combinations of tactics.

Limitations & Future Work

Limitations

- ▶ Small-scale pilot study
- ▶ Limited compute budget
- ▶ Small Hard_Lean evaluation
- ▶ Pseudo-Lean transformations are imperfect

Future Work

- ▶ Larger multilingual datasets
- ▶ Deeper proof-search analysis
- ▶ Stronger theorem proving models

Main Perspective

This work is an initial step toward understanding **why multilingual training helps theorem proving.**

Thank you!